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jtleeek / genstats

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Branch: gh-pagesgenstats / inst / doc / 01_10_exploratory-analysis.R

jtleeek12 days ago big update

1 contributor

163 lines (127 sloc)5.02 KB

RawBlameHistory

```
1  ## ----global_palette, results = 'asis'-----
2  rm(list=ls())
3  tropical= c('darkorange', 'dodgerblue', 'hotpink', 'limegreen', 'yellow')
4  palette(tropical)
5
6  ## ----global_options,warning=FALSE,message=FALSE-----
7  ## see ch. 10 Hooks of Xie's knitr book
8  library(knitr)
9  knit_hooks$set(setPch = function(before, options, envir) {
10    if(before) par(pch = 19)
11  })
12  opts_chunk$set(setPch = TRUE)
13
14  ## ----global_plot,warning=FALSE, message=FALSE-----
15  knitr::opts_chunk$set(fig.width=5, fig.height=5, size="footnotesize",
16                        warning=FALSE, message=FALSE)
17  knitr::knit_hooks$set(small.mar = function(before, options, envir) {
18    if (before) graphics::par(mar = c(5,5,1.5,1))
19  })
20
21  ## ----load_hidden, echo=FALSE, results="hide", warning=FALSE-----
22  suppressPackageStartupMessages({
23    library(devtools)
24    library(Biobase)
25    library(RSkittleBrewer)
26    library(gplots)
27    library(dplyr)
28    library(AnnotationDbi)
29  })
30
31  ## ----load-----
32  library(gplots)
33  library(devtools)
34  library(Biobase)
35  library(RSkittleBrewer)
36  library(org.Hs.eg.db)
37  library(AnnotationDbi)
38
39  ## ----install_packages, eval=FALSE-----
40  # install.packages(c("devtools","gplots"))
41  # source("http://www.bioconductor.org/biocLite.R")
42  # biocLite(c("Biobase","org.Hs.eg.db","AnnotationDbi"))
43  # biocLite("alyssafranze/RSkittleBrewer")
44
45  ## ----pretty, eval=FALSE-----
46  # library(RSkittleBrewer)
47  # # Make the colors pretty
48  # trop = RSkittleBrewer("tropical")
49  # palette(trop)
50  # par(pch=19)
51
52  ## ----load_data-----
53  con = url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bodymap_eset.RData")
54  load(file=con)
55  close(con)
56  bm = bodymap.eset
57  pdata=pData(bm)
58  edata=exprs(bm)
59  fdata = fData(bm)
60  ls()
61
62  ## ----tables-----
63  table(pdata$gender)
```

```

64 table(pdata$gender,pdata$race)
65
66 ## ----summary-----
67 summary(edata)
68
69 ## ----missing-----
70 # Use option useNA to include NA's in table
71 table(pdata$age,useNA="ifany")
72
73 # is.na checks for NA values
74 table(is.na(pdata$age))
75
76 # Check for other common missing names
77 sum(pdata$age==" ")
78
79 # Check genomic data for NAs
80 sum(is.na(edata))
81
82 # Make the distribution of NA's by genes
83 gene_na = rowSums(is.na(edata))
84 table(gene_na)
85
86 # Make the distribution of NA's by samples
87 sample_na = rowSums(is.na(edata))
88 table(sample_na)
89
90
91 ## ----dimensions-----
92 dim(fdata)
93 dim(pdata)
94 dim(edata)
95
96 ## ----boxplot-----
97 boxplot(log2(edata+1),col=2,range=0)
98
99 ## ----histograms-----
100 par(mfrow=c(1,2))
101 hist(log2(edata[,1]+1),col=2)
102 hist(log2(edata[,2]+1),col=2)
103
104 ## ----densities-----
105 plot(density(log2(edata[,1]+1)),col=2)
106 lines(density(log2(edata[,2]+1)),col=3)
107
108 ## -----
109 qqplot(log2(edata[,1]+1), log2(edata[,2]+1),col=3)
110
111 ## -----
112 mm = log2(edata[,1]+1) - log2(edata[,2]+1)
113 aa = log2(edata[,1]+1) + log2(edata[,2]+1)
114 plot(aa,mm,col=2)
115
116 ## -----
117 edata = as.data.frame(edata)
118 filt_edata = filter(edata,rowMeans(edata) > 1)
119 boxplot(as.matrix(log2(filt_edata+1)),col=2)
120
121 ## -----
122 aeid = as.character(fdata[,1])
123 chr = AnnotationDbi::select(org.Hs.eg.db,keys=aeid,keytype="ENSEMBL",columns="CHR")
124 head(chr)
125
126 ## -----
127 dim(chr)
128 dim(edata)
129 # Take non-duplicated chromosomes
130 chr = chr[!duplicated(chr[,1]),]
131
132 # Confirm that the annotation still is in the right order
133 all(chr[,1] == rownames(edata))
134
135 # Select the chromosome Y samples
136 edatay = dplyr::filter(edata,chr$CHR=="Y")
137
138 # Males have Y chromosome expression as expected
139 boxplot(colSums(edatay) ~ pdata$gender)
140 points(colSums(edatay) ~ jitter(as.numeric(pdata$gender)),
141        col=as.numeric(pdata$gender),
142        pch=19)
143

```

```
144
145 ## -----
146 ematrix = as.matrix(edata)[rowMeans(edata) > 10000,]
147 heatmap(ematrix)
148
149 ## -----
150 colramp = colorRampPalette(c(3,"white",2))(9)
151 heatmap(ematrix,col=colramp)
152
153 ## -----
154 heatmap(ematrix,col=colramp,Rowv=NA,Colv=NA)
155
156 ## -----
157 heatmap.2(ematrix,col=colramp,Rowv=NA,Colv=NA,
158           dendrogram="none", scale="row",trace="none")
159
160 ## ----session_info-----
161 devtools::session_info()
162
```

